

Elio Govea

Senior Software Engineer · [Vaisala](#)

Twelve years of C++ in networking and embedded Linux. ICPC World Finalist.

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Profile

Software engineer at the seam of networking, embedded Linux, and large-scale infrastructure. Currently at [Vaisala](#) in Vantaa, building [AviMet](#) — an aviation weather platform Vaisala has deployed across **170+ countries** (2,500+ aviation systems delivered). Before that, four-plus years at [F-Secure](#) in Helsinki and Poznań on the Sense SDK — C++ home-router security shipped to **200+ service-provider partners** across x86, ARM, and MIPS, then leading CDN cost optimization for it.

Trained in competitive programming at the [Universidad de Pinar del Río](#) in Cuba. Represented the university at three ICPC Latin American Finals (silver in 2016) and at the [2018 ICPC World Finals](#) in Beijing — Honourable Mention (top 140 teams worldwide, advanced from ~50,000+ regional teams).

Highlights

12+

years building C++ for networking and embedded Linux

2018

ICPC World Finals · Honourable Mention

80%

reduction in monthly CDN spend for Sense SDK deployments

Experience

[Vaisala](#) · Senior Software Engineer

02/2025 – present

Vantaa, Finland · [AviMet](#) aviation weather platform

- Drove AviMet's Linux platform expansion — added **Alma Linux** support across the C++/Qt/CMake build matrix and toolchain, and implemented Linux-side license enforcement, integrating the platform with Vaisala's existing licensing system.
- Built tooling for a new cloud entitlements system to enable customer activations
- Tracked down and fixed long-dormant bugs that started surfacing under newer usage patterns and platforms — memory leaks and corner-case crashes that had been latent for years.
- Stack: C++/Qt on Windows and Linux, CMake; Python/Bash/PowerShell scripting; JavaScript/TypeScript web tooling.

F-Secure · Senior Developer

05/2023 – 02/2025

Helsinki, Finland

- Designed CDN cost-reduction strategy for the Sense SDK — cut **monthly CDN spend by 80%** with minimal impact on end-users.
 - Root-caused the cost to a router-side behavior driving excessive CDN traffic. The natural fix — a router software update — would have taken months to reach deployed devices given the rigorous home-router release and testing cycle.
 - Designed a backend-side mitigation that neutralized the problem server-side, no router changes required — shipping in days instead of months.
 - Followed up with a router-software fix for future releases, closing the underlying cause permanently.
- Co-designed and implemented cloud-driven dynamic traffic-rule enforcement in the Sense SDK — enabled F-Secure to push new security policy to deployed routers from the cloud without firmware updates.

F-Secure · Developer

12/2021 – 04/2023

Helsinki, Finland · Sense SDK — embedded by 200+ service-provider partners

- **Built Sense SDK** — F-Secure's C++ SDK for home-router security: URL filtering and parental controls (forcing safe-mode browser engines), traffic-statistics collection, device recognition and fingerprinting; across IPv4/IPv6, DHCP, TCP/UDP, DNS, HTTP/HTTPS using Linux networking interfaces (netlink, netfilter, iptables, nfqueue, nflog, conntrack, BPF) and integrated with cloud back-end services.
- **Shipped on x86, ARM, and MIPS targets** — designed and optimized the cross-platform build system (CMake + GNU Make) for fast, reproducible builds across toolchains, plus SDK integrations for reference platforms to streamline demos and future integrations.
- **Designed Jenkins / AWS CI pipelines** — enforced code quality through coverage, static, and dynamic analysis; produced builds for multiple architectures and reference platforms.
- **Built testing infrastructure in Python and Robot Framework**; implemented recovery mechanisms for OS- and cloud-service-level errors.

F-Secure · Software Developer

09/2020 – 11/2021

Poznań, Poland

- Contributed to the core C++ of the Sense SDK.
- Optimized a notifications system for security events.
- Developed embedded Linux clients to consume cloud services.
- Added endpoints to existing REST APIs.
- Improved internal C++ interfaces and data structures.
- Fixed bugs in existing products.
- Added new interfaces using Unix domain sockets for communication between clients and services.

Pinar del Río, Cuba

- Built real-time C++/Qt applications for vibration-signal processing and IMU-based object position tracking — embedded sensor input (Arduino, serial), Linux, OpenGL visualization.
- Assisted in teaching: prepared lectures (C++, Python, Arduino, Linux, Raspberry Pi), ran labs, designed and graded assignments.
- ICPC judge for the Caribbean region and team coach — coached a team to the ICPC Latin American Finals.

Projects

DNS library

Parsing and generating DNS messages from scratch.

Library to parse and emit DNS messages, with example resolvers demonstrating common queries.

[C++](#)[Networking](#)[DNS](#)

Router-network Docker testbed

Client → router → server, all in containers.

Docker setup that simulates a small home network — clients behind a router-as-gateway talking to remote servers — for testing network applications end-to-end without real hardware.

[Docker](#)[Networking](#)[Linux](#)

C++ project template

Reusable scaffold for cross-platform C++ work.

Docker environment with common tools, CMake presets for multiple configurations, and CI pipelines for code coverage and documentation.

[C++](#)[CMake](#)[Docker](#)[CI](#)

Suffix automaton visualization

Interactive walk-through of a classic string algorithm.

TypeScript and D3 tool that illustrates each step of building a suffix automaton.

[Live demo ↗](#)

[TypeScript](#)[D3](#)[Algorithms](#)

Template metaprogramming examples

Types and constants computed at compile time.

Worked C++ examples that build up and transform types and values at compile time. Each example focuses on a single idea.

[C++](#)[Templates](#)[constexpr](#)

Competitive programming archive

The algorithms I leaned on, and the problems I solved with them.

Two companion repositories. The first is my personal reference of algorithms and data structures — the kind of notebook ICPC teams build up over years. The second is the running set of solutions I wrote against online-judge problems.

[Reference](#) · [Solutions](#)

[C++](#)[Algorithms](#)[Data structures](#)[ICPC](#)

Baseball Limits 2D

The "limits" of baseball performance, two stats at a time.

What does "best ever" look like when you compare two baseball stats at once? In one dimension it's a ranking; in two it's a Pareto frontier — the players no one strictly outperforms in both. Interactive viewer over 150 years of MLB data.

[Live demo ↗](#)

[JavaScript](#)[Python](#)[Data viz](#)[Pareto frontier](#)[Baseball](#)

Education

Universidad de Pinar del Río · B.Sc. Telecommunications and Electronics Engineering

09/2012 – 07/2017

Pinar del Río, Cuba

- GPA **4.95 / 5.00**
- ICPC World Finalist

Achievements

- **Honourable Mention** — 2018 ACM ICPC World Finals, Beijing (*top 140 teams worldwide, advanced from ~50,000+ regional teams*)
- **2nd Place (Silver Medal)** — 2016 ACM ICPC Latin American Finals
- **4th Place** — 2015 ACM ICPC Latin American Finals
- **Round 3 (Top 1000)** — 2019 Google Code Jam (*of ~50,000 entrants — top ~2%*)

Skills

LANGUAGES

C C++ Python Bash TypeScript / JavaScript

BUILD & TOOLING

CMake GNU Make Git Docker LXC Jenkins AWS CI

SYSTEMS

Linux Buildroot OpenWrt Alma Linux Windows

CONCEPTS

Algorithms Data Structures Networking Embedded systems Cross-platform builds CI/CD
Performance optimization

PROTOCOLS

IP DHCP UDP TCP DNS HTTP TLS QUIC

Languages

- **Spanish** — Native
- **English** — Full professional proficiency